An exact period-three Chambolle–Pock orbit at $\theta = 1/8$ below the quadratic threshold

Abstract

We give a rational four-dimensional convex projection example for the literal primal-first Chambolle–Pock map with extrapolation $\theta = 1/8$. The stepsize product is 3, strictly below the quadratic benchmark $16/5$, yet an explicitly initialized orbit has exact period three. A positive definite rational Gram matrix and two triangles specify the entire instance. All projection optimality conditions are verified by finite tables of rational inner products. The example disproves the proposed universal convex extension of that benchmark, not a published convergence theorem. It is bounded, its ordinary averages converge to a saddle point, and it does not determine the sharp universal threshold. The construction transfers isometrically to ordinary Euclidean space. A standalone exact-arithmetic verifier accompanies the proof. No novelty claim is made.

Keywords: Chambolle–Pock iteration; period-three orbit; convex projection; rational Gram certificate; quadratic threshold counterexample

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1 The recurrence and the claim being tested

Let H be a real Hilbert space. For a proper lower-semicontinuous convex function f, write

$$\operatorname{prox}_{\tau f}(u) = \operatorname*{argmin}_{v \in H} \left\{ f(v) + \frac{\|v-u\|^2}{2\tau} \right\}, \qquad \tau > 0.$$

The primal-first map considered here is precisely

$$\begin{aligned} x_{n+1} &= \operatorname{prox}_{\tau f}(x_n - \tau L^* y_n), \\ y_{n+1} &= \operatorname{prox}_{\sigma h}\big(y_n + \sigma L((1+\theta)x_{n+1} - \theta x_n)\big). \end{aligned} \tag{1}$$

Here $h = g^*$ in the usual saddle notation. There is no outer relaxation, correction step, or alternative initialization rule. The KKT set is

$$Z = \{(x,y) : -L^*y \in \partial f(x),\ Lx \in \partial h(y)\}.$$

At $\theta = 1/8$, the expression $4/(1+2\theta)$ equals $16/5$. The assertion tested in this note is that *every* proper lower-semicontinuous convex instance with $Z \neq \varnothing$ and $\tau\sigma\|L\|^2 < 16/5$ has convergent full iterates under (1). We disprove that assertion. We do not attribute it to a published theorem. The recurrence itself originates in the primal–dual framework of Chambolle and Pock [1].

2 Exact problem data and theorem

In coordinate space $\mathbb{R}^4$, let

$$Q = \begin{pmatrix} 476 & -238 & 266 & -14 \\ -238 & 476 & -252 & 266 \\ 266 & -252 & 190 & -95 \\ -14 & 266 & -95 & 190 \end{pmatrix}, \qquad \langle u, v \rangle_Q = u^{\mathsf{T}} Q v. \tag{2}$$

Let $e_1, \dots, e_4$ denote the coordinate vectors and set

$$a_1 = e_1, \quad a_2 = e_2, \quad a_3 = -e_1 - e_2, \qquad b_1 = e_3, \quad b_2 = e_4, \quad b_3 = -e_3 - e_4.$$

Define the triangles and functions

$$C = \operatorname{conv}\{a_1, a_2, a_3\}, \qquad D = \operatorname{conv}\{b_1, b_2, b_3\}, \qquad f = \iota_C, \quad h = \iota_D,$$

where ι_E is zero on E and $+\infty$ elsewhere. All conjugates and subdifferentials are taken in the Q inner product. In particular,

$$g(z) = h^*(z) = \max_{1 \le j \le 3} \langle z, b_j \rangle_Q, \qquad g^* = h.$$

Take the same Hilbert space for primal and dual variables, with

$$L = I_H, \qquad \tau = 3, \qquad \sigma = 1, \qquad \theta = 1/8.$$

Theorem 1. *The matrix Q is positive definite. The functions f, h are proper lower-semicontinuous convex, $(0,0) \in Z$, and $\|L\| = 1$. The initialized recurrence*

$$x_{n+1} = P_C^Q(x_n - 3y_n), \qquad y_{n+1} = P_D^Q\left(y_n + \frac{9x_{n+1} - x_n}{8}\right), \qquad (x_0, y_0) = (a_3, b_3) \tag{3}$$

has the exact orbit

$$(x_{3k}, y_{3k}) = (a_3, b_3), \quad (x_{3k+1}, y_{3k+1}) = (a_1, b_1), \quad (x_{3k+2}, y_{3k+2}) = (a_2, b_2) \quad (k \ge 0).$$

It has least period three and does not converge, even weakly, although

$$\tau\sigma\|L\|^2 = 3 < \frac{16}{5}.$$

3 Positive definiteness and the saddle point

The leading principal minors of Q, in order, are

$$476, \qquad 169932, \qquad 286552, \qquad 362404.$$

All are positive, proving positive definiteness by Sylvester's criterion. For a second exact certificate, $Q = T\Delta T^{\mathsf{T}}$, where

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1/2 & 1 & 0 & 0 \\ 19/34 & -1/3 & 1 & 0 \\ -1/34 & 37/51 & -1/2 & 1 \end{pmatrix}, \qquad \Delta = \operatorname{diag}\left(476, 357, \frac{86}{51}, \frac{43}{34}\right).$$

Thus (2) defines a genuine Hilbert norm. Both triangles are nonempty compact convex sets, so their indicators have all the stated properties. Their vertices sum to zero, giving $0 \in C \cap D$. For any closed convex set E and $v \in E$, its normal cone is

$$N_E^Q(v) = \{r : \langle r, w - v\rangle_Q \le 0 \text{ for every } w \in E\}.$$

In particular $0 \in N_C^Q(0)$ and $0 \in N_D^Q(0)$. Since $\partial \iota_E = N_E^Q$, this proves the KKT assertion. Equivalently, the saddle function

$$\mathcal{L}(x, y) = f(x) + \langle x, y\rangle_Q - h(y)$$

has a saddle point at $(0, 0)$: its value is zero for $x = 0, y \in D$ and for $x \in C, y = 0$. Finally $I_H^* = I_H$ and $\|I_H\| = 1$, because both spaces use exactly the same norm.

4 All three projection certificates

For a nonempty closed convex set E, the projection criterion is

$$v = P_E^Q(u) \quad \Longleftrightarrow \quad v \in E, \quad \langle u - v, w - v\rangle_Q \le 0 \quad (w \in E). \tag{4}$$

For completeness, expanding $\|u-w\|_Q^2$ about v shows that the displayed inequality implies minimality. Conversely, minimality applied to $v + t(w - v)$ and the limit $t \downarrow 0$ gives the inequality. Positive definiteness gives uniqueness. For a triangle it suffices to check its three vertices, since the left side is affine in w.

Interpret subscripts cyclically, so $a_0 = a_3$ and $b_0 = b_3$. For the transition to phase $i \in \{1, 2, 3\}$ define the *raw projection residuals*

$$r_i = a_{i-1} - 3b_{i-1} - a_i, \qquad s_i = b_{i-1} + \frac{9a_i - a_{i-1}}{8} - b_i.$$

The explicit residuals are

i	r_i^{T}				s_i^{T}			
1	-2	-1	3	3	5/4	1/8	-2	-1
2	1	-1	-3	0	$-1/8$	9/8	1	-1
3	1	2	0	-3	$-9/8$	$-5/4$	1	2

Direct multiplication by the rational matrix Q gives every required vertex inequality:

$$\begin{aligned}\big[\langle r_i, a_j - a_i\rangle_Q\big]_{i,j=1}^{3} &= \begin{pmatrix} 0 & 0 & -126 \\ -126 & 0 & 0 \\ 0 & -126 & 0 \end{pmatrix}, \\ \big[\langle s_i, b_j - b_i\rangle_Q\big]_{i,j=1}^{3} &= \begin{pmatrix} 0 & -1/4 & -191/4 \\ -191/4 & 0 & -1/4 \\ -1/4 & -191/4 & 0 \end{pmatrix}.\end{aligned} \tag{5}$$

Every entry is nonpositive. Thus (4) proves

$$P_C^Q(a_{i-1} - 3b_{i-1}) = a_i, \qquad P_D^Q\left(b_{i-1} + \frac{9a_i - a_{i-1}}{8}\right) = b_i \quad (i = 1, 2, 3).$$

The first primal update is therefore literally from (a_3, b_3) to a_1, and its dual update uses that newly obtained a_1. Induction now proves the entire initialized orbit, rather than only a cyclic compatibility condition. Note that the proximal subgradient for f is $r_i/3$, whose table is the first table in (5) divided by three; its nonzero entries are -42. The dual subgradient is s_i because $\sigma = 1$. These conventions describe the same normals.

Completion of the theorem. All instance assumptions were checked above, and the projection certificates establish the recurrence. The three states are distinct, so the sequence has least period three. Its constant phase subsequences have three different limits. In finite dimensions weak and strong convergence coincide, so neither kind of full-sequence convergence is possible. The product inequality follows from $\|L\| = 1$ and $16/5 - 3 = 1/5 > 0$. ∎

5 Ordinary Euclidean realization and scope

Let $S = Q^{1/2}$ be the symmetric positive definite square root. The linear map $u \mapsto Su$ is an isometry from $(\mathbb{R}^4, \langle\cdot,\cdot\rangle_Q)$ to ordinary Euclidean $\mathbb{R}^4$. Put

$$\widehat{C} = SC, \quad \widehat{D} = SD, \quad \widehat{f} = \iota_{\widehat{C}}, \quad \widehat{h} = \iota_{\widehat{D}}, \quad \widehat{L} = I.$$

The change of variables $v = Sw$ in the projection minimization gives

$$P_{SE}(Su) = SP_E^Q(u).$$

Consequently $(\widehat{x}_n, \widehat{y}_n) = (Sx_n, Sy_n)$ satisfies the ordinary Euclidean version of (3), with the same stepsizes and extrapolation, and starts at (Sa_3, Sb_3). The transformed coupling is $SIS^{-1} = I$, its adjoint and norm are still I and 1, and $(0, 0)$ is still a saddle point. No change of algorithm or non-Euclidean exception is responsible for the cycle.

The functions are indicators of triangles, so this is a genuinely nonlinear projection example rather than a quadratic spectral example. It defeats a universal point-convergence extension of the quadratic benchmark at $\theta = 1/8$. It does not invalidate a published

theorem, nor the separate $\theta = 1/4$ result: the extrapolation parameter changes the map. It supplies an obstruction at product 3, but does not prove that 3 or any other value is the sharp universal threshold.

In particular this orbit is bounded. Indeed it takes just three values, and

$$(a_1, b_1) + (a_2, b_2) + (a_3, b_3) = (0, 0).$$

Every consecutive block of three states therefore sums to zero, and its ordinary Cesàro averages converge to $(0, 0)$. The example does not disprove boundedness or averaged convergence.

6 Reproducibility

The accompanying file `verify.py` requires Python 3 and SymPy. Run `python3 verify.py`. It reconstructs the literal update inputs from the initialized phase list, checks exact principal minors and the LDL^{T} factorization, computes both complete normal tables, and verifies each projection inequality and the strict product comparison using integers and rational numbers only. It also checks twelve successive updates via the projection criterion. No floating-point tolerance or numerical semidefinite certificate enters either the proof or the verifier.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.